

Understanding JSON vs Protocol Buffers (Protobuf)

Lecture Notes for First-Year Students

Protocol Buffers Wire Types

Every field in a Protobuf message is encoded using a **field number** and a **wire type**. The wire type tells the receiver how to interpret the bytes that follow.

Wire Type	Value	Meaning	Common Protobuf Types
Varint	0	Variable-length integer	int32, int64, uint32, uint64, sint32, sint64, bool, enum
64-bit	1	Always 8 bytes	fixed64, sfixed64, double
Length-delimited	2	Length followed by data	string, bytes, embedded messages, repeated (packed) fields
Start group	3	Deprecated	Not used in modern Protobuf
End group	4	Deprecated	Not used in modern Protobuf
32-bit	5	Always 4 bytes	fixed32, sfixed32, float

Examples of Wire Types

int32 orderId = 101

Wire Type 0 (Varint) because integers are encoded efficiently.

bool isPaid = true

Wire Type 0 because booleans are stored as 0 or 1.

string customerName = "Fayaz"

Wire Type 2 because strings are variable length. The message contains the string length followed by ASCII/UTF-8 bytes.

double salary = 55000.75

Wire Type 1 because doubles always occupy 8 bytes.

float temperature = 36.5

Wire Type 5 because floats always occupy 4 bytes.

bytes image

Wire Type 2 because raw binary data is length-delimited.

Embedded Address object

Wire Type 2 because nested messages are prefixed with their length.

Memory Trick

0 = Variable-sized numbers (Varint)

1 = 64-bit fixed size

2 = Anything with variable length (strings, bytes, nested objects)

3 & 4 = Deprecated groups

5 = 32-bit fixed size

How the First Byte is Formed

Formula: **(Field Number << 3) | Wire Type**

Example: customerName has field number 2 and wire type 2.

$(2 \ll 3) = 16$. $16 | 2 = 18$ decimal = $0x12$.

So the first byte is **12**.

Why This Matters

When the receiver reads the first byte, it immediately knows two things:

1. Which field is being sent.
2. How to decode the bytes that follow.

This combination of field numbers and wire types is one of the reasons Protocol Buffers are compact and extremely fast.